Tables

1 $\frac{5}{36}$

2

a Sample Space $= \{0, 1, 2, 3, 4, 5\}$

b $\frac{1}{6}$

3

a

Spinner	2	3	4
7	9	10	11
9	11	12	13
12	14	15	16

b **i** $\frac{2}{9}$ **ii** $\frac{2}{9}$ **iii** $\frac{3}{9}$

4

a

	1	2	3	4	5	6
1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4	4, 1	4, 2,	4, 3	4, 4	4, 5	4, 6
5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

b

i $\frac{1}{18}$ **ii** $\frac{1}{36}$ **iii** $\frac{1}{6}$ **iv** $\frac{1}{2}$

5

a

	W	X	Y	Z
1	1, W	1, X	1, Y	1, Z
2	2, W	2, X	2, Y	2, Z
3	3, W	3, X	3, Y	3, Z
4	4, W	4, X	4, Y	4, Z
5	5, W	5, X	5, Y	5, Z
6	6, W	6, X	6, Y	6, Z

b 24

c **i** $\frac{1}{8}$ **ii** $\frac{1}{6}$ **iii** $\frac{11}{12}$

6

a

	8	2	3	5
1	8	2	3	5
2	16	4	6	10
3	24	6	9	15
4	32	8	12	20
5	40	10	15	25
6	48	12	18	30

b $\frac{1}{4}$

c $\frac{1}{12}$

d $\frac{2}{3}$

7

a

	Cyan (C)	Pink (P)	White (W)
Black (B)	C, B	P, B	W, B
Grey (G)	C, G	P, G	W, G
Red (R)	C, R	P, R	W, R
Yellow (Y)	C, Y	P, Y	W, Y

b 12

c $\frac{1}{4}$

d $\frac{1}{3}$

8

a

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	25	41	4	70
Liberal (last election)	16	73	1	90
Greens (last election)	9	26	5	40
Total	50	140	10	200

b $\frac{8}{45}$

c $\frac{1}{2}$

d $\frac{57}{80}$

9

a

Spinner	2	4	5
3	5	7	8
7	9	11	12

b i 1 ii $\frac{5}{6}$ iii $\frac{2}{5}$ iv 1

Tree diagrams

10 $\frac{1}{9}$

11

a SR, SL, SC, GV, GM, GC

b $\frac{1}{6}$

c $\frac{1}{3}$

12

a $\{2, 3, 5, 7, 11, 13, 17, 19\}$

b i $\frac{1}{8}$ ii $\frac{1}{4}$

13

a $23, 24, 32, 34, 42, 43$

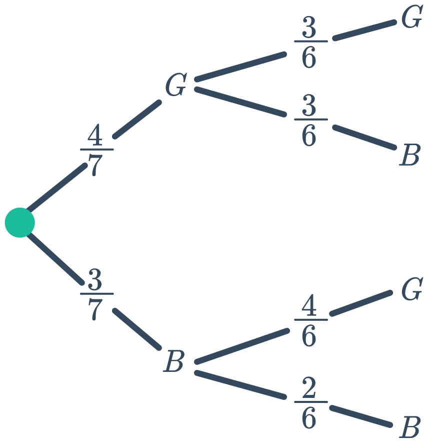
b i $\frac{2}{3}$ ii $\frac{1}{3}$ iii $\frac{1}{3}$

14

a 0.42 b 0.88

15

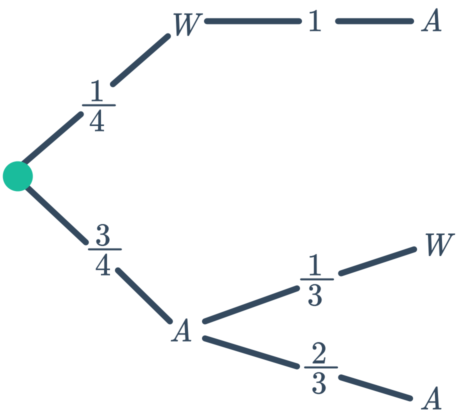
a



b **i** $\frac{3}{7}$ **ii** $\frac{2}{7}$

16

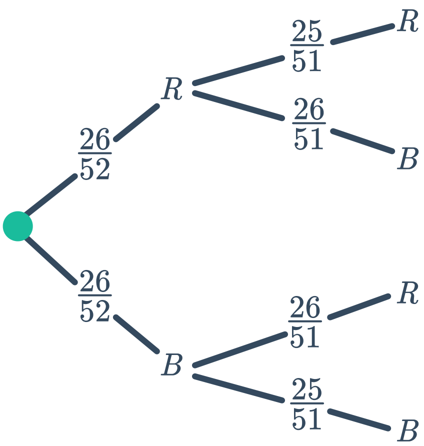
a



b **i** $\frac{3}{4}$ **ii** 1

17

a



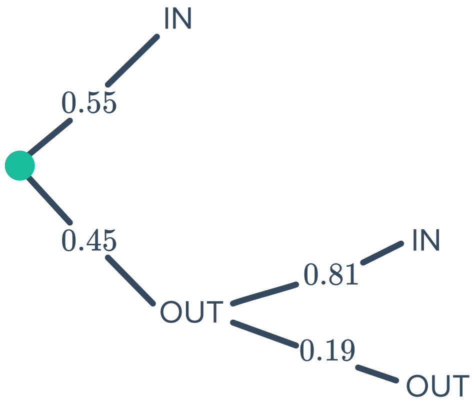
b $\frac{13}{51}$

c

i Yes **ii** No **iii** No **iv** No

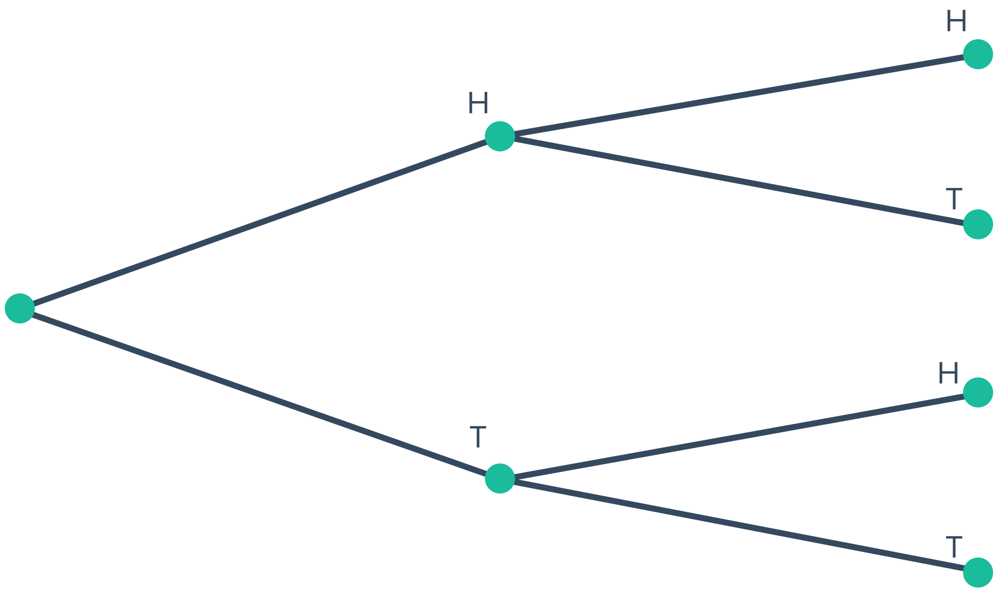
v No

18 **a**



b **i** 0.45 **ii** 0.0855

19 **a**

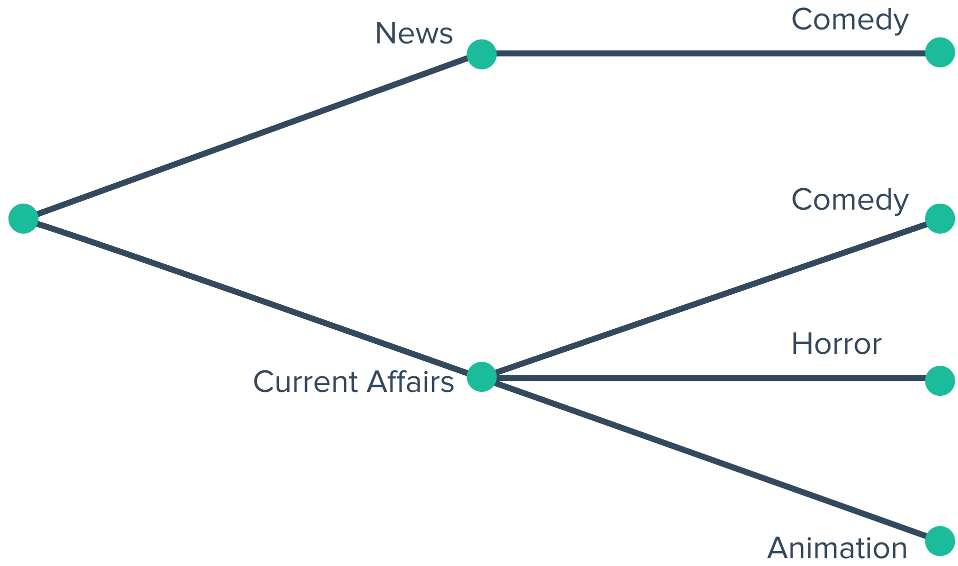


b

i $\frac{2}{4}$ **ii** $\frac{1}{4}$ **iii** $\frac{1}{4}$ **iv** $\frac{2}{4}$

20

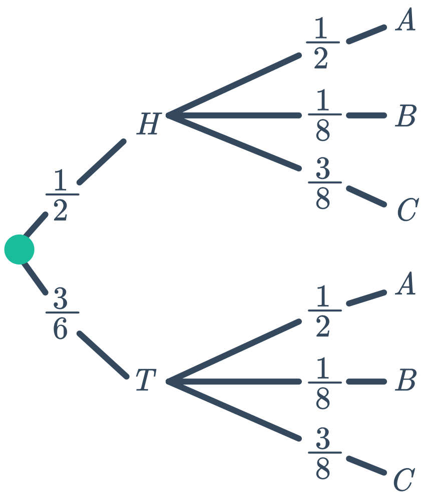
a



b $\frac{2}{3}$

21

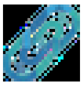
a



b **i** $\frac{3}{16}$ **ii** $\frac{11}{16}$

22 $\frac{1}{7776}$

23



a

i $\frac{1}{9}$ ii $\frac{1}{27}$ iii $\frac{19}{27}$

b $\mathrm{P(Winning\ at\ least\ one\ of\ two\ games)} = \frac{5}{9}, \mathrm{P(Winning\ at\ least\ one\ of\ three\ games)} = \frac{19}{27}$

therefore Neville has a better chance at winning at least one of three games.

24

a

i $\frac{1}{2}$ ii $\frac{1}{4}$ iii $\frac{1}{8}$

b $\frac{1}{2}$

c $\frac{1}{16}$

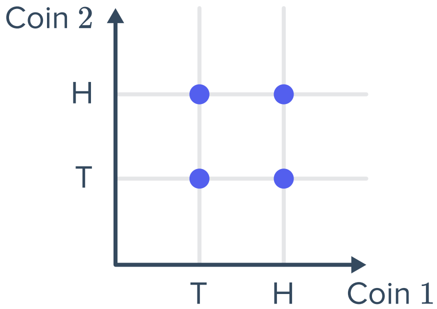
25

a $\frac{1}{2}$ **b** $\frac{1}{32}$

2D Grids

26

a

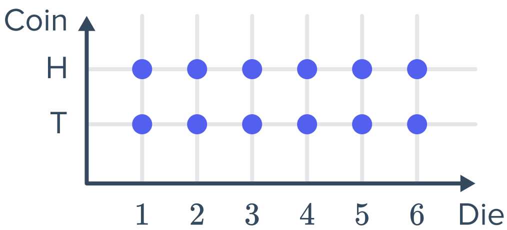


b 4

c $\frac{1}{2}$

27

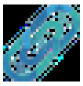
a



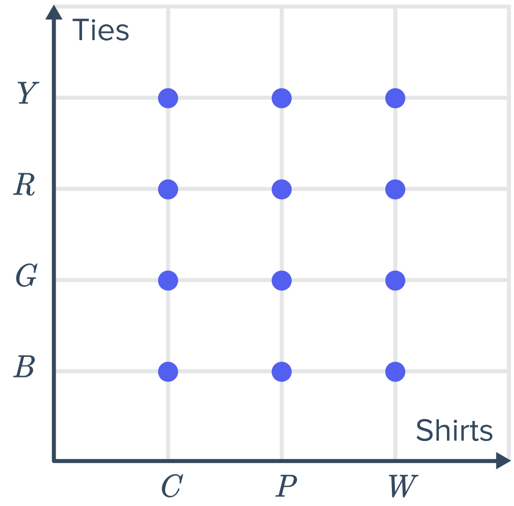
b 12

28 a

	C	P	W
B	C, B	P, B	W, B
G	C, G	P, G	G, W
R	C, R	P, R	W, R
Y	C, Y	P, Y	W, Y



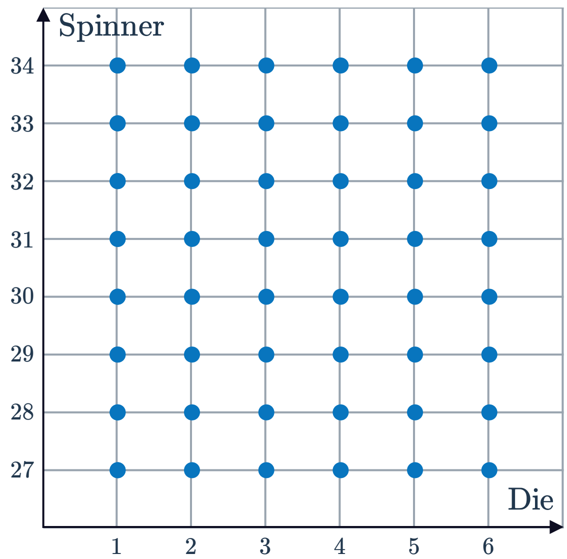
b



c 12

29

a



b 48

c $\frac{1}{4}$